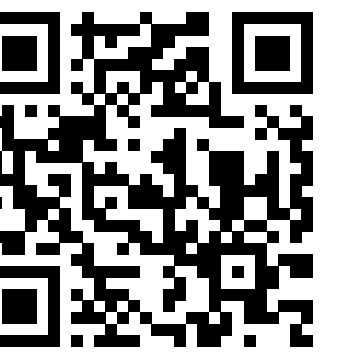
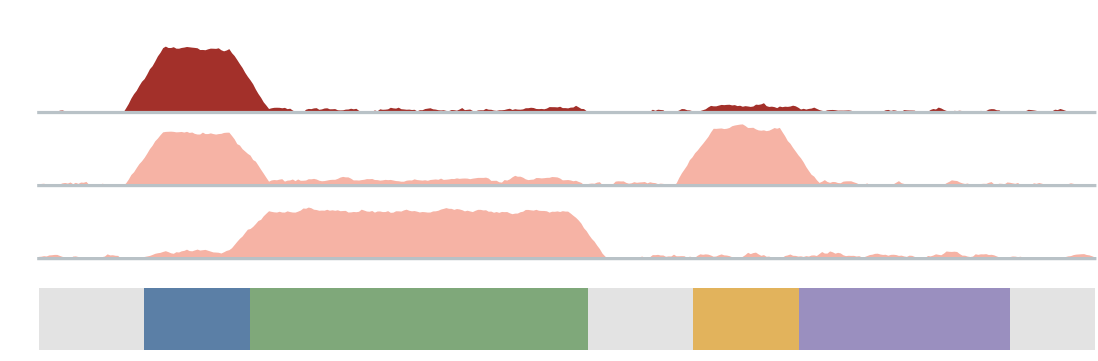




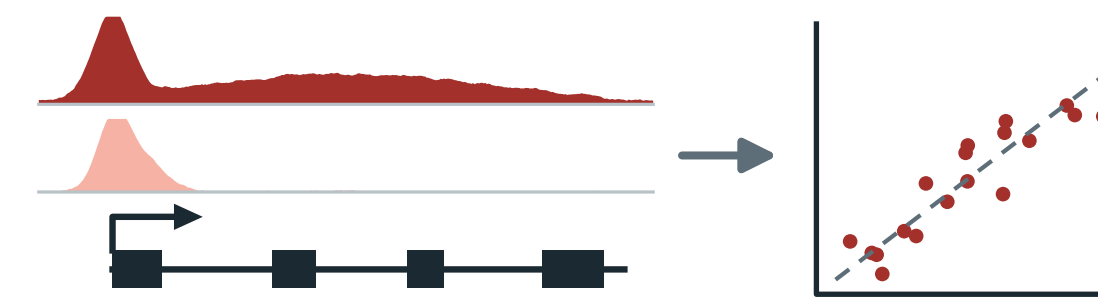
The problem

- Epigenomic data form a 3-D tensor: tens to hundreds of assays $\times$ thousands of cell types $\times$ genomic positions.
- Profiling every assay in every cell type costs too much, so most experiments are missing.
- Imputation predicts these missing experiments [1,2].
- Imputed data are often cleaner than observed data, so imputation also denoises performed experiments [1].
- Experimental covariates matter: in the ENCODE Imputation Challenge, a train-test shift let a simple average-activity baseline beat most entrants [2].

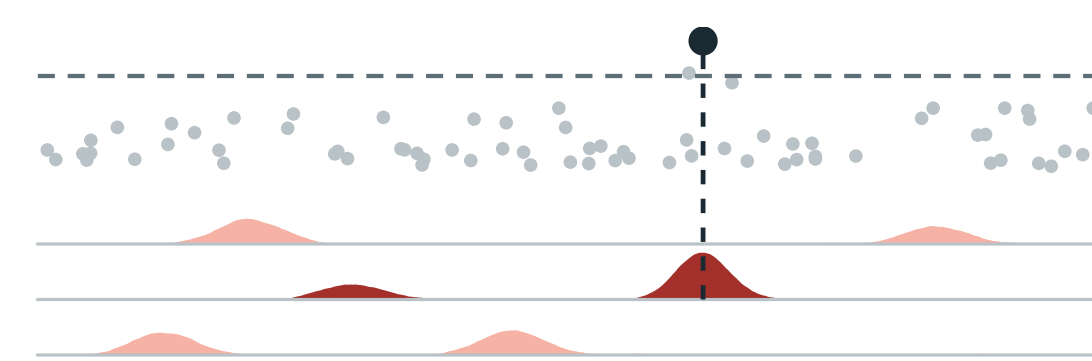
Downstream applications



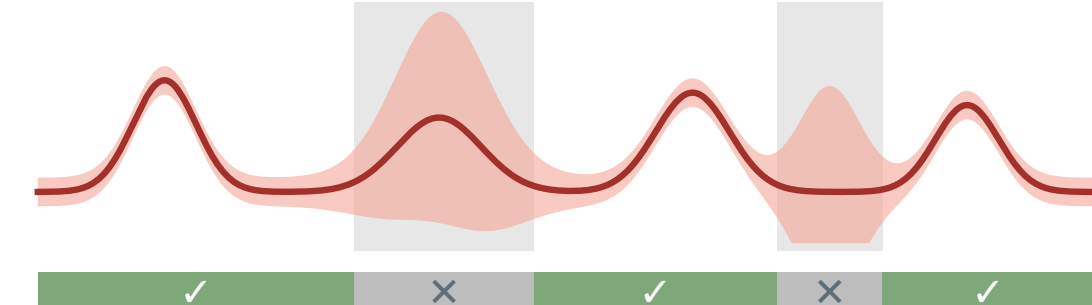
Annotate chromatin states in every cell type [1].



Predict gene expression from marks around each gene [1,3].

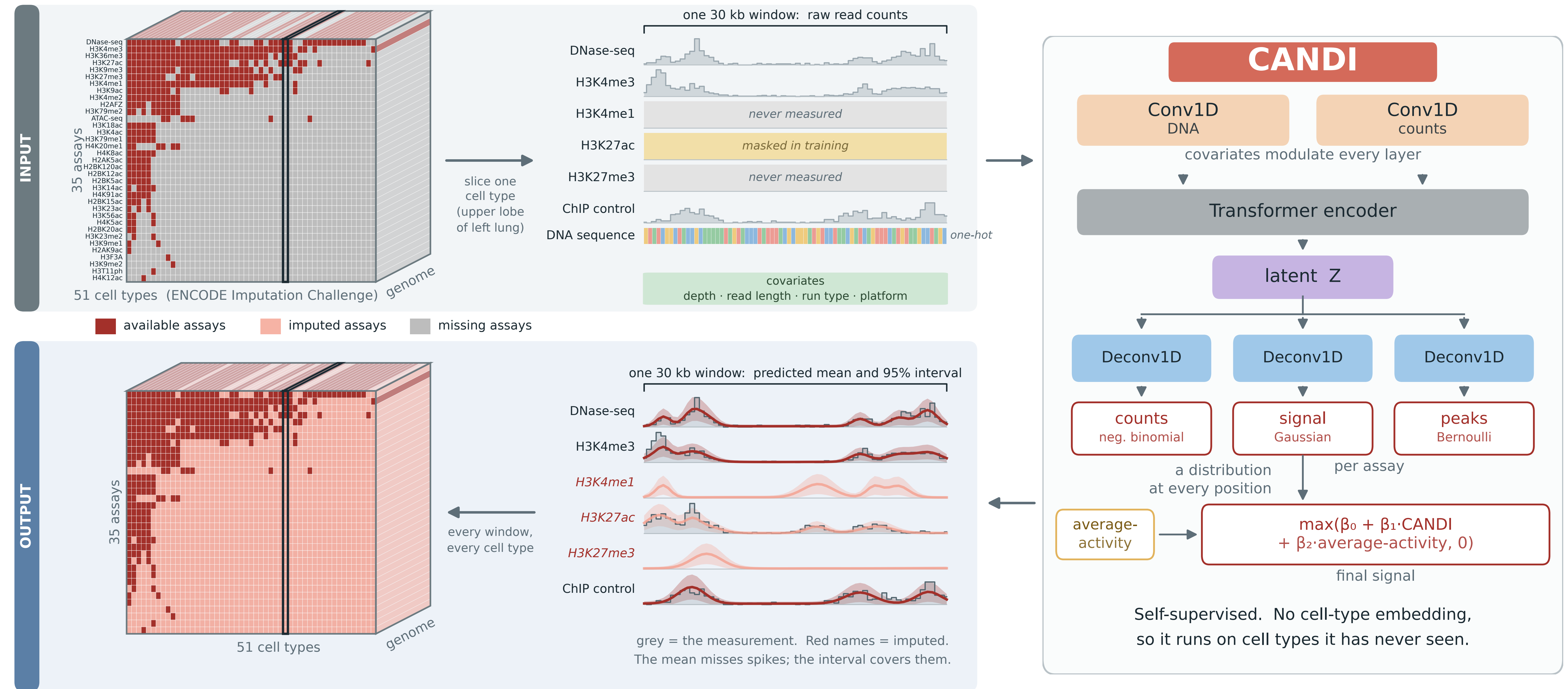


Link disease-associated variants to relevant cell types [1].



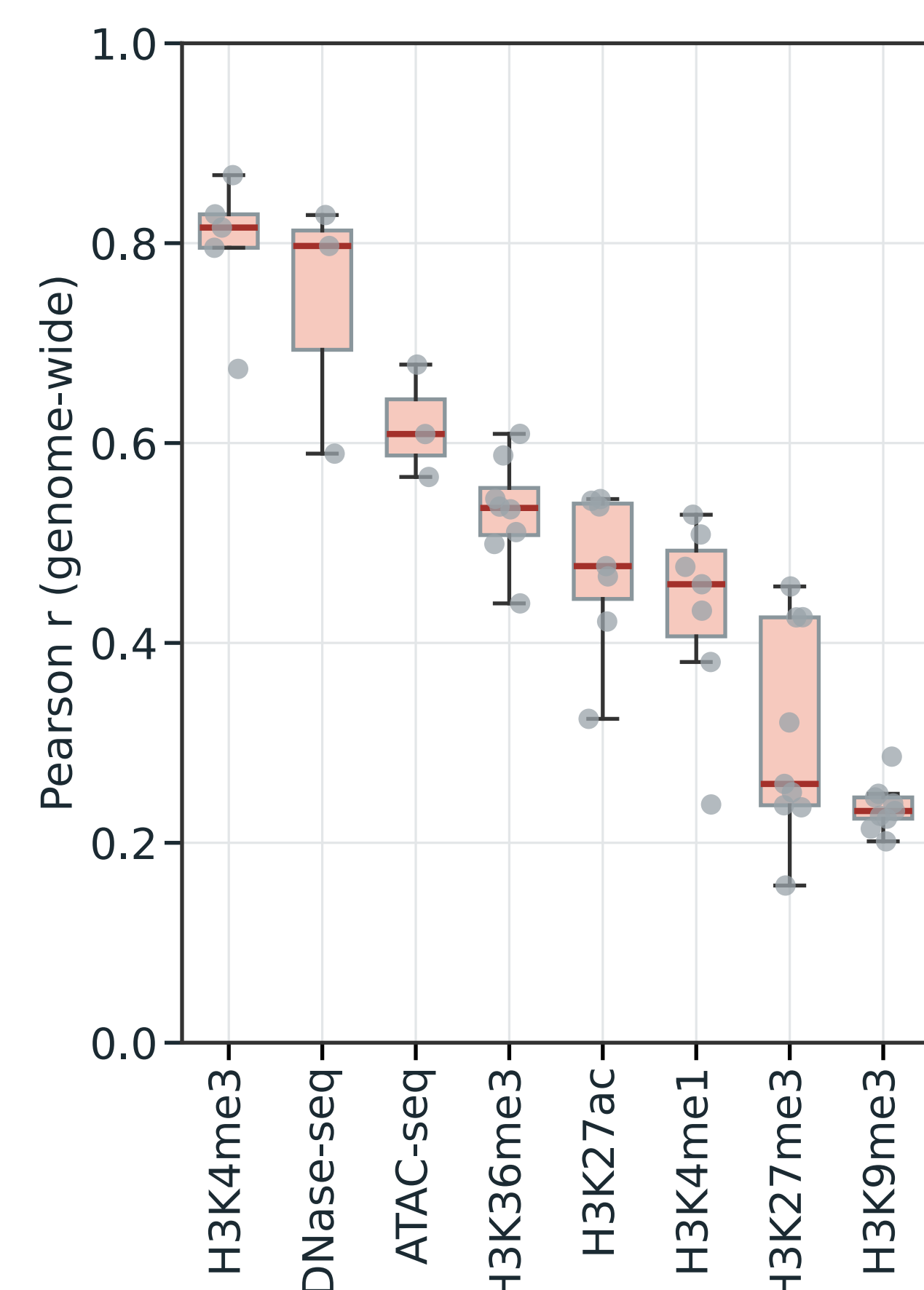
Keep confident predictions; set aside uncertain ones [3].

CANDI schematic workflow

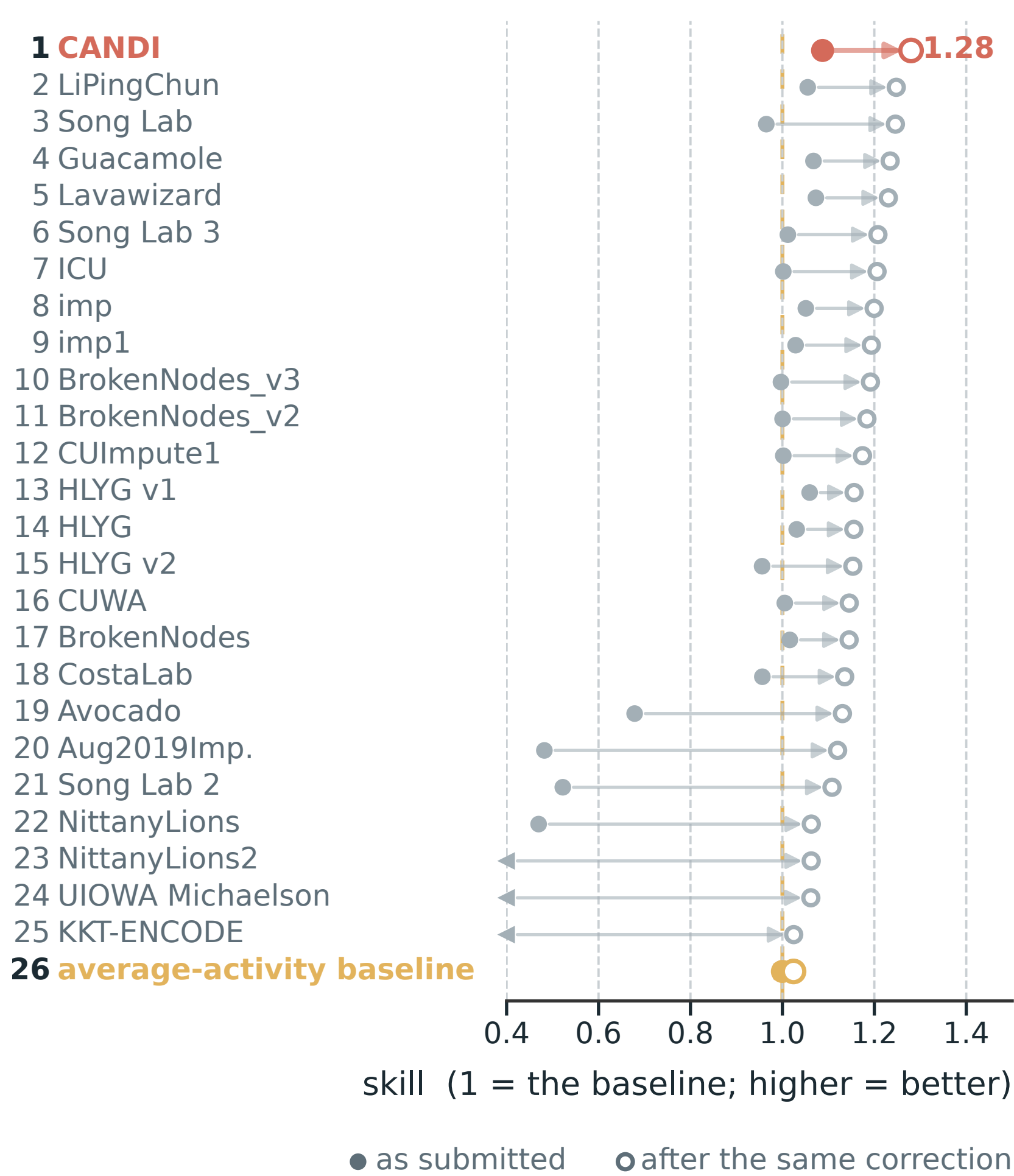


Top: for each 30 kb locus, CANDI takes observed read counts with the ChIP-seq control, four covariates per assay and the DNA sequence; grey assays were never measured, yellow ones are masked. **Bottom:** each output is a probability distribution, here a mean and 95% interval per assay; window by window, they fill every missing thread of the tensor.

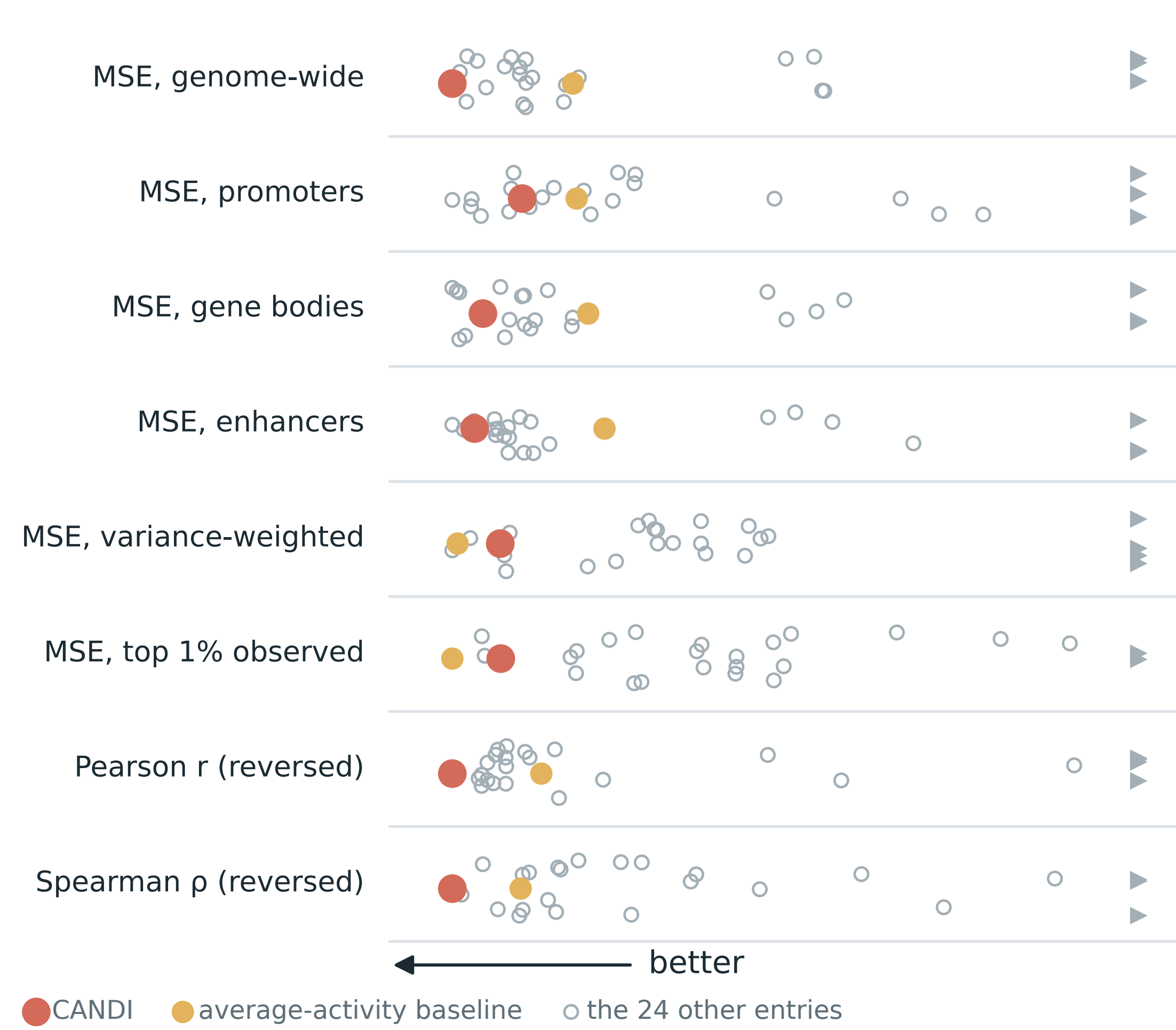
CANDI fixes the train-test covariate shift in the ENCODE Imputation Challenge



A Genome-wide Pearson r of CANDI against each of the 51 blind-test experiments (points), by assay.

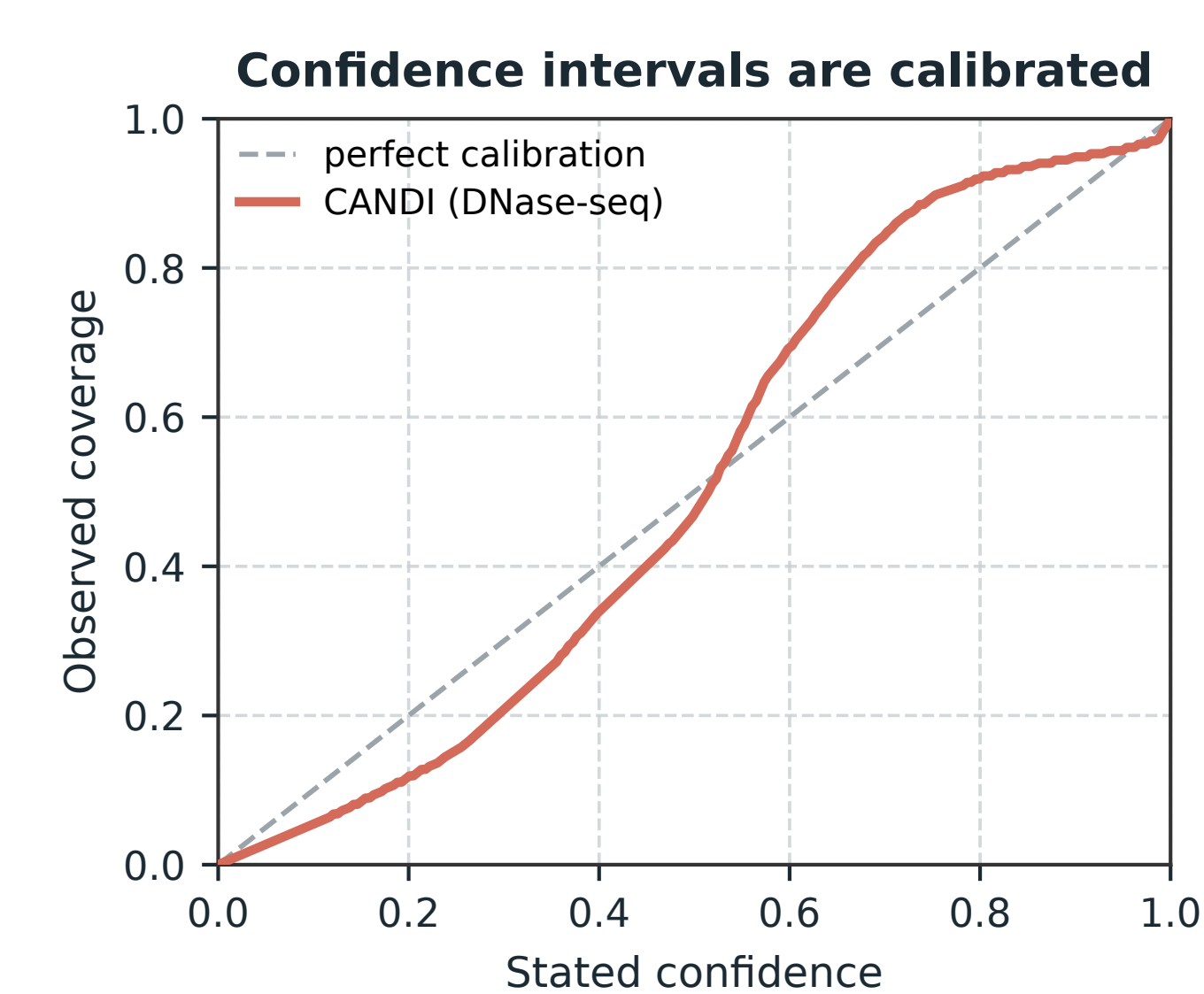


B Skill as submitted (filled) and after the same correction for every entry (open); 1 is the baseline. CANDI is first both ways.

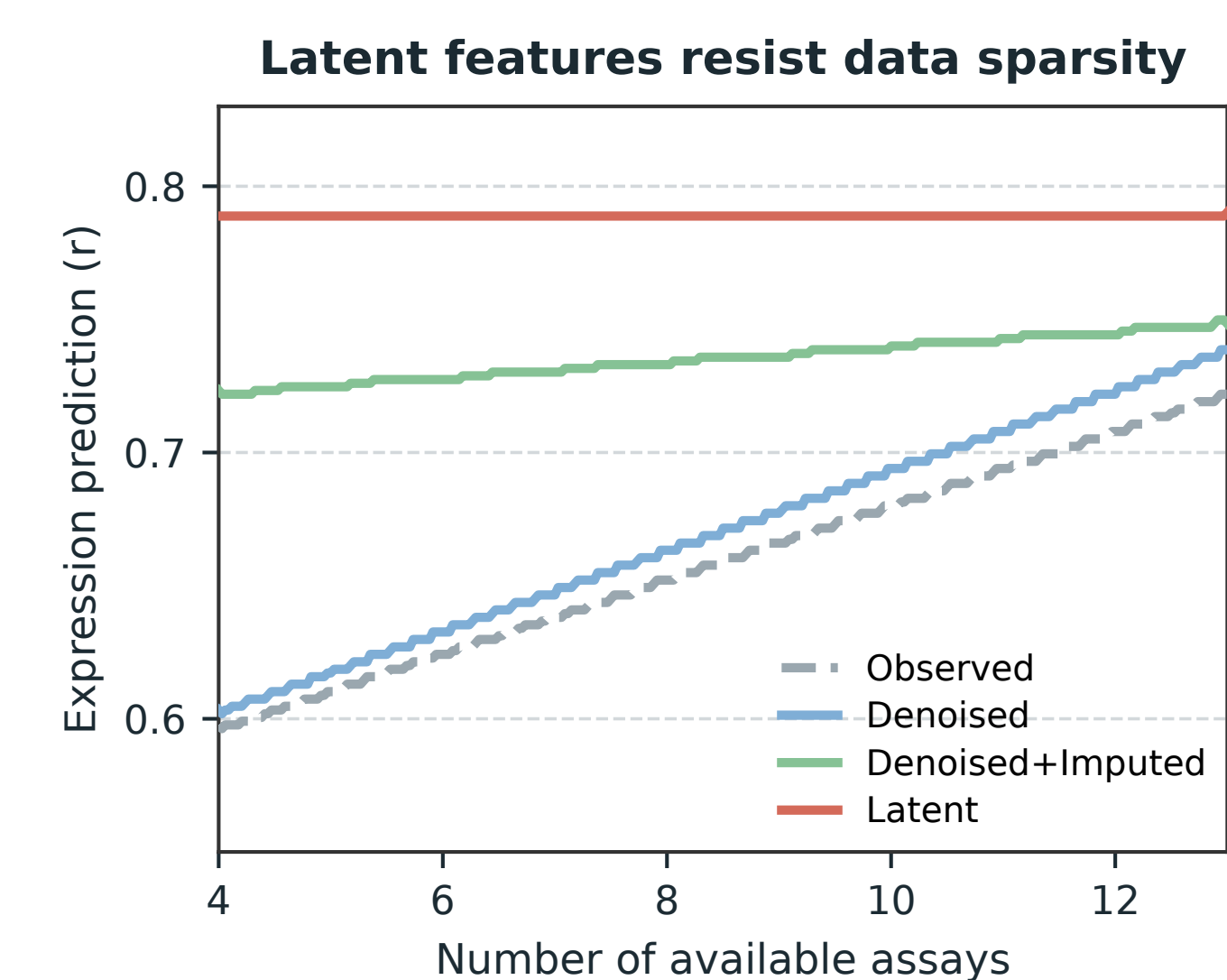


C Eight of the nine measures relative to the baseline, median over 51 experiments. CANDI is first on three and top 8 on the rest.

Downstream utility of CANDI



D Confidence calibration for DNase-seq (DND-41): predicted confidence level against the fraction of observed data within that interval.



E Gene expression prediction as input assays are removed. Denoised+Imputed and Latent settings are robust to input sparsity.

[1] Ernst J, Kellis M. Large-scale imputation of epigenomic datasets for systematic annotation of diverse human tissues. *Nat Biotechnol* 33:364–376 (2015).

[2] Schreiber J, Boix C, et al. The ENCODE Imputation Challenge: a critical assessment of methods for cross-cell type imputation of epigenomic profiles. *Genome Biol* 24:79 (2023).

[3] Foroozandeh M, Diab AR, Libbrecht M. CANDI: self-supervised, confidence-aware denoising imputation of genomic data. *bioRxiv* 2025.01.23.634626 (2025).

Panels A–C: blind-test scoring from the Libbrecht lab's EIC reproduction; *baseline* is the challenge's average-activity track. CANDI's output layer is fitted on training experiments only; the correction in B is fitted on chr22 of each blind-test experiment and scored on the other 22 chromosomes.